

A candidate counterexample to Question 6.4 in Extending Thomassen’s conjecture to directed graphs

github.com/graph-theory-AI — machine-generated candidate writeup

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Abstract

The anti-directed C_4 is not Eulerian-avoidable: Eulerian oriented graphs can have arbitrarily large minimum out-degree while having no anti-directed- C_4 -free subdigraph of minimum out-degree 2. This candidate disproof was generated by GPT-6 Astra in a single model pass for the Graph Theory AI project. It was selected from the campaign because the model marked it `would_publish: true`; it has not been checked by the adversarial referee, a human mathematician, or a novelty search.

1 The problem and the claimed result

This document concerns [Question 6.4](#) from *Extending Thomassen’s conjecture to directed graphs*.

1.1 Problem statement

Question. Is every orientation of C_4 Eulerian-avoidable?

1.2 Catalog context

Restricting to Eulerian digraphs (where each vertex has equal out- and in-degree) may make avoidability questions more tractable. The authors ask whether the specific case of C_4 , which is open in the general setting, becomes resolvable in the Eulerian host digraph setting.

1.3 Claimed result

Model verdict: disproved (confidence: high).

The anti-directed C_4 is not Eulerian-avoidable: Eulerian oriented graphs can have arbitrarily large minimum out-degree while having no anti-directed- C_4 -free subdigraph of minimum out-degree 2.

The model’s own caveat: The counterexamples have very large degree spread; this does not classify the other orientations of C_4 .

2 Model writeup

The text below is the model output verbatim, apart from moving its machine-readable verdict block into the summary above and shifting Markdown heading levels for this document. Mechanical TeX defects and equation tags were normalized where needed for compilation.

2.1 A counterexample for the anti-directed orientation

Let $F = \vec{K}_{2,2}$, with arcs

$$u \rightarrow x, \quad u \rightarrow y, \quad v \rightarrow x, \quad v \rightarrow y.$$

All digraphs below are finite and simple, and copies need not be induced.

Theorem. For every integer $d \geq 2$, there is a strongly connected Eulerian oriented graph D_d with

$$\delta^+(D_d) = d$$

such that no nonempty subdigraph of D_d has both minimum out-degree at least 2 and no copy of F .

Thus the answer to the question is **no**.

The construction combines a small terminal bottleneck with a long sequence of regular, high-girth bipartite layers. Avoiding F bounds the number of surviving vertices immediately before the bottleneck. The high-girth layers then force these numbers to decrease strictly when traced backwards, eventually giving a contradiction. In particular, deleting vertices cannot circumvent the obstruction.

2.1.1 1. A small-target growth lemma

Lemma 1. Let Q be a bipartite graph with parts X, Y and girth greater than $2M$. Suppose that nonempty sets $P \subseteq X$ and $R \subseteq Y$ satisfy

$$|R| \leq M$$

and every vertex of P has at least two neighbors in R . Then

$$|P| \leq |R| - 1.$$

Proof. Any cycle in $Q[P, R]$ would have length at most $2|R| \leq 2M$, because a bipartite cycle uses equally many distinct vertices from each part. Consequently, $Q[P, R]$ is a forest. Therefore

$$2|P| \leq e(Q[P, R]) \leq |P| + |R| - 1,$$

which proves the claim. $\square$

We will use regular bipartite graphs of arbitrarily large prescribed girth. A fully explicit, self-contained construction is given in Section 4.

2.1.2 2. Construction of the Eulerian host

Fix $d \geq 2$, and put

$$M = \binom{d}{2}, \quad L = M + 1.$$

Choose a simple d -regular bipartite graph Q , with n vertices in each part, whose girth is greater than $2M$.

Take disjoint vertex sets

$$B, A_1, A_2, \dots, A_L, \quad |B| = d, \quad |A_i| = n.$$

Define D_d by the following arcs:

1. All arcs from B to A_1 .

2. For each $1 \leq i < L$, a copy of Q between A_i and A_{i+1} , with every edge directed from A_i to A_{i+1} .
3. All arcs from A_L to B .

There are no other arcs. Schematically,

$$B \longrightarrow A_1 \xrightarrow{Q} A_2 \xrightarrow{Q} \cdots \xrightarrow{Q} A_L \longrightarrow B.$$

Every vertex of B has in-degree and out-degree n . Every vertex of every A_i has in-degree and out-degree d . Since $n \geq d$,

$$d_{D_d}^-(z) = d_{D_d}^+(z) \quad \text{for all } z, \quad \delta^+(D_d) = d.$$

Also, $L \geq 2$, so this is an oriented graph: there are no antiparallel pairs.

The digraph is strongly connected. Every vertex in an A_i can reach B by following the layers forwards. Conversely, tracing predecessors backwards through the regular layers shows that every vertex in an A_i is reachable from A_1 , and hence from every vertex of B . Thus the construction satisfies even the convention that an Eulerian digraph must be connected.

2.1.3 3. No F -free subdigraph can have minimum out-degree 2

Suppose, for a contradiction, that a nonempty subdigraph $H \subseteq D_d$ is F -free and satisfies

$$\delta^+(H) \geq 2.$$

Set

$$T = V(H) \cap B, \quad S_i = V(H) \cap A_i.$$

All these sets are nonempty. Indeed, every arc goes to the next class in the cyclic sequence

$$B, A_1, \dots, A_L, B.$$

Starting at any vertex of H and repeatedly following an outgoing arc therefore visits every class.

2.1.3.1 The final layer is small For each $z \in S_L$, choose two distinct out-neighbors of z in H . Both belong to T ; denote their unordered pair by P_z .

If two distinct vertices $z, z' \in S_L$ had $P_z = P_{z'}$, these two sources and their two common heads would form a copy of $\vec{K}_{2,2}$. Hence the pairs P_z are all distinct, and

$$|S_L| \leq \binom{|T|}{2} \leq \binom{d}{2} = M. \quad (3.1)$$

This is the elementary pair-counting obstruction from the supplied lead, now applied at a designated final layer.

2.1.3.2 Every preceding layer is strictly smaller Suppose $1 \leq i < L$ and $|S_{i+1}| \leq M$. Every vertex of S_i has at least two out-neighbors in S_{i+1} , since these are its only possible out-neighbors in H .

The underlying graph between A_i and A_{i+1} is a copy of Q . Lemma 1 therefore gives

$$|S_i| \leq |S_{i+1}| - 1. \quad (3.2)$$

Starting from (3.1) and applying (3.2) backwards through the $L - 1 = M$ intermediate transitions yields

$$|S_1| \leq |S_L| - (L - 1) \leq M - M = 0,$$

contradicting the nonemptiness of S_1 .

This proves the theorem. Notice that H was not assumed to be spanning or Eulerian.

If an Eulerian-avoidability function d_F existed, choosing

$$d \geq \max\{2, d_F(2)\}$$

and applying its defining property to D_d would contradict the theorem. Therefore $\vec{K}_{2,2}$ is not Eulerian-avoidable. $\square$

2.1.4 4. An explicit construction of the high-girth layers

For completeness, the required regular bipartite graph can be constructed without invoking any external existence theorem.

Lemma 2. For every $d \geq 2$ and $g \geq 2$, one can explicitly construct a finite simple d -regular bipartite graph of girth greater than g .

Proof. Let W consist of the empty word and all words of length at most g over the alphabet $[d]$ in which consecutive letters differ.

For each $j \in [d]$, define a permutation σ_j of W as follows:

- If a word ends in j , delete its final letter.
- Otherwise, if its length is less than g , append j .
- Otherwise, fix it.

Each σ_j is an involution. We use right actions for permutation multiplication.

If

$$j_1 j_2 \cdots j_\ell, \quad 1 \leq \ell \leq g,$$

has consecutive letters distinct, then

$$\emptyset \sigma_{j_1} \sigma_{j_2} \cdots \sigma_{j_\ell} = j_1 j_2 \cdots j_\ell.$$

In particular,

$$\sigma_{j_1} \sigma_{j_2} \cdots \sigma_{j_\ell} \neq 1. \quad (4.1)$$

Let $\Gamma = \text{Sym}(W)$. Form a bipartite graph Q whose two parts are copies Γ_0, Γ_1 of Γ , with edges

$$\pi_0 (\pi \sigma_j)_1 \quad (\pi \in \Gamma, j \in [d]).$$

The generators $\sigma_1, \dots, \sigma_d$ are distinct, so Q is simple and d -regular. Give each displayed edge color j . Each color class is a perfect matching.

Suppose Q had a simple cycle of length $\ell \leq g$. Reading its edge colors in order gives a sequence $j_1, \dots, j_\ell$ with consecutive colors distinct: using the same color twice consecutively would immediately traverse the same matching edge backwards. Since every σ_j is an involution, returning to the starting vertex gives

$$\sigma_{j_1} \sigma_{j_2} \cdots \sigma_{j_\ell} = 1,$$

contrary to (4.1). Hence Q has girth greater than g . $\square$

For the counterexample, take $g = 2M = d(d - 1)$. This gives a completely specified family. More explicitly, one may use

$$|W| = 1 + d \sum_{r=0}^{d(d-1)-1} (d-1)^r, \quad n = |W|!,$$

so that

$$|V(D_d)| = d + \left(\binom{d}{2} + 1 \right) n.$$

These sizes are deliberately unoptimized.

2.1.5 5. Scope and a generalization

The same construction proves a broader statement:

Corollary. For all $s, t \geq 2$, the one-directed complete bipartite digraph $\vec{K}_{s,t}$, directed from its s -vertex part to its t -vertex part, is not Eulerian-avoidable.

To see this, fix $d \geq t$, use target minimum out-degree $k = t$, and replace the parameters by

$$M = (s-1) \binom{d}{t}, \quad L = M + 1.$$

In a $\vec{K}_{s,t}$ -free subdigraph, choosing a t -set of out-neighbors for each final-layer vertex assigns any particular t -set to at most $s - 1$ sources. Thus the final layer again has at most M vertices. Since $t \geq 2$, Lemma 1 again forces strict growth forwards through every small intermediate layer, giving the same contradiction.

For the original question, only $s = t = 2$ is needed. The crucial distinction from complete cyclic blow-ups is that the intermediate high-girth layers cannot support a small 2-out core. This removes the vertex-deletion escape available in the supplied attempt and gives a counterexample to Eulerian-avoidability itself, not merely to a spanning version.

A Provenance and review status

UNREFEREED MODEL OUTPUT. This document was selected solely because GPT-6 Astra labelled its own result `would_publish: true` and returned `proved` or `disproved`. It has not passed the adversarial LLM referee used for the earlier Sol campaign, has not been checked by a human mathematician, and has not been checked for novelty. Treat every mathematical and bibliographic claim as unverified.

Catalog id	2510.11311__04
Catalog entry	Question 6.4
Source corpus	arxiv
Campaign leg	selected second attempts (attacks_retry)
Source paper / entry	Extending Thomassen's conjecture to directed graphs
Model verdict	disproved (confidence: high)
Selection criterion	<code>would_publish: true</code> and verdict <code>proved</code> or <code>disproved</code>
Model	<code>gpt-6-astra</code> , reasoning effort max, mode=pro, flex service tier
Original artifact	<code>attacks_retry/2510.11311__04/output.md</code>
Independent review	None
arXiv	arXiv:2510.11311

Previous attempt	attacks/2510.11311__04: gpt-5.6-sol verdict partial
Model's caveats	The counterexamples have very large degree spread; this does not classify the other orientations of C4.

The generated Markdown and LaTeX sources for this PDF are stored under `to_review_astra/src/2510.11311__04/`.